The Norwegian fund is always in the public eye, and the actions and words of the responsible officials at the Ministry of Finance and NBIM are closely scrutinized by the press. The outcry at the active losses was special, partly because the losses occurred alongside contracting economic growth and the uncovering of other financial excesses—Norway, despite its stolid reputation, was not spared the effects of the financial crisis. Also, Norwegian parliamentarians are financially sophisticated; they have to be, as they are ultimately responsible for the nest egg of a nation. (Ask the average member of Congress what's the difference between active and passive management and he'll think you're referring to different exercise routines.)

The Ministry of Finance wisely did not immediately yank NBIM's active mandate despite the public outcry. Instead, it commissioned a four-month, in-depth study of the fund. William Goetzmann (Yale University), Stephen Schaefer (London Business School), and I were tasked with evaluating NBIM's historical performance, assessing its strategic plans for active management and its risk budgeting process and describing and evaluating investing strategies that played to Norway's comparative advantages as a large, patient, and transparent investor. Released at the end of 2009 and presented in January 2010 at a public meeting introduced by the King of Norway, the report was quickly dubbed the "Professors' Report," or more conventionally, Ang, Goetzmann, and Schaefer (2009).

We found that a major part of the fund's active returns before, during, and after the financial crisis could be explained by systematic factors. It was right for Norway to collect these risk premiums in the long run, which compensated Norway for losses during bad times like the financial crisis. Many of these factors could be collected more cheaply by *passive management*. The report recommended that NBIM's active mandate should not be rescinded but that the Norwegian fund should go beyond equities and bonds in its asset allocation: Norway should adopt a top–down approach of factor investing, especially for *dynamic factors*.

2. Factors Really Matter

2.1. WHAT IS A FACTOR?

Factors are investment styles that deliver high returns over the long run. The risk premiums don't come for free, however, as factors can underperform in the short run ("*bad times*"). Factor losses are associated with bad economic outcomes, like times of high inflation and slow economic growth. Equities and bonds are examples of factors whose risk premiums are obtained by simply buying assets (*long-only* positions), hence they are *static* factors.[3] Other

[3] Of course, you would want to rebalance equity and positions over time, as chapter 4 advocates, to maintain constant risk exposures to equity and bond market risk. You can rebalance over both static factors and dynamic factors.